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MULTI-TAP MECHANICAL SHEAR BOLT INSULATION PIERCING CONNECTORS

Abstract

A connector includes a conductor block with a run line opening, a run insulation piercing member extending into the run line opening, a tap line opening, and a tap insulation piercing member extending into the tap line opening. A run cap extends over the run line opening. A run bolt extends through the run cap into the run line opening. The run bolt includes a first threaded shank and a first shearable head. The first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit. A tap cap extends over the tap line opening. A tap bolt extends through the tap cap into the tap line opening. The tap bolt includes a second threaded shank and a second shearable head. The second shearable head is configured to disconnect from the second threaded shank on reaching a second torque limit.

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Background/Summary

RELATED APPLICATION(S) [0001] This application is based on U.S. Provisional Application Ser. No. 63/561,944, filed Mar. 6, 2024 and U.S. Provisional Application Ser. No. 63/730,704, filed Dec. 11, 2024, the disclosure of which is incorporated herein by reference in its entirety and to which priority is claimed.

FIELD

[0002] The present disclosure relates to electrical connectors used to connect a main or run conductor with one or more tap conductors.

BACKGROUND

[0003] Electrical connectors exist in various forms including terminals, splices and run and taps. Tap connectors can be used to establish an electrical connection between a continuous run or main power conductor to a tap or branch conductor. Existing compression connectors are typically adapted to receive a branch or tap conductor, to engage a continuous run conductor, and to be compressed by means of a crimping tool to achieve the desired connection.

[0004] Compression connectors can include insulation piercing teeth that will pierce the non-conductive, outer insulation layer of a conductor to contact the inner core. The core can be a solid or stranded wire made from an electrically conductive material. Use of insulation piercing teeth can reduce or eliminate the need to strip conductors prior to making a connection and eliminate the need to secure connections with a joining material such as solder.

SUMMARY

[0005] Certain configurations are directed to an electrical connector for connecting a plurality of conductors. The connector includes a conductor block having a body with a run line opening, a run insulation piercing member extending into the run line opening, a tap line opening, and a tap insulation piercing member extending into the tap line opening. A run cap extends over the run line opening. A run bolt extends through the run cap into the run line opening. The run bolt includes a first threaded shank and a first shearable head. The first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit. A tap cap extends over the tap line opening. A tap bolt extends through the tap cap into the tap line opening. The tap bolt includes a second threaded shank and a second shearable head. The second shearable head is configured to disconnect from the second threaded shank on reaching a second torque limit.

[0006] Certain configurations are directed to an electrical connector for connecting a plurality of conductors. The connector includes a conductor block having a body with a run line opening, a run insulation piercing member extending into the run line opening, a run guide slot extending from the run line opening, a tap line opening, a tap insulation piercing member extending into the tap line opening, and a tap guide slot extending from the tap line opening. A run cap is connected to the conductor block via the run guide slot. A run bolt extends through the run cap into the run line opening. The run bolt includes a first threaded shank and a first shearable head. The first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit. A tap cap is connected to the conductor block via the tap guide slot. A tap bolt extends through the tap cap into the tap line opening. The tap bolt includes a second threaded shank and a second shearable head. The second shearable head is configured to disconnect from the second threaded shank on reaching a second torque limit.

[0007] Certain implementations are directed to a method of connecting a plurality of electrical

conductors. A tap conductor is inserted into a tap line opening of a conductor block. A tap bolt is rotated to engage the tap conductor. The tap conductor is driven into a tap insulation piercing member extending into the tap line opening. The tap bolt is rotated until a first torque limit is reached and a head of the tap bolt is sheared off. A run conductor is inserted into a run line opening of the conductor block. A run bolt is rotated to engage the run conductor. The run conductor is driven into a run insulation piercing member extending into the run line opening. The run bolt is rotated until a second torque limit is reached and a head of the run bolt is sheared off.

[0008] Certain configurations are directed to an electrical connector for connecting a plurality of conductors. A conductor block includes a body with a run line opening, a run insulation piercing member extending into the run line opening, a tap line opening, and a tap insulation piercing member extending into the tap line opening. A run cap extends over the run line opening. A run bolt extends through the run cap into the run line opening. The run bolt includes a first threaded shank and a first shearable head. The first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit. A tap bolt extends through the conductor block into the tap line opening.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The aspects and features of various exemplary embodiments will be more apparent from the description of those exemplary embodiments taken with reference to the accompanying drawings.

[0010] FIG. 1 is a perspective view of an exemplary configuration of a multi-tap electrical connector.

[0011] FIG. 2 is a front view of the connector of FIG. 1.

[0012] FIG. 3 is a top view of the connector of FIG. 1.

[0013] FIG. 4 is a front, exploded view of the connector of FIG. 1.

[0014] FIG. 5 is a perspective view of another exemplary configuration of a multi-tap electrical connector.

[0015] FIG. 6 is a front, exploded view of the connector of FIG. 5.

[0016] FIG. 7 is a perspective view of another exemplary configuration of a multi-tap electrical connector.

[0017] FIG. 8 is a bottom perspective view of the connector of FIG. 7.

[0018] FIG. 9 is a front, exploded view of the connector of 7.

[0019] FIG. 10 is a front view of another exemplary configuration of a multi-tap electrical connector.

[0020] FIG. 11 is a front view of another exemplary configuration of a multi-tap electrical connector.

[0021] FIG. 12 is a front view of another exemplary configuration of a multi-tap electrical connector.

[0022] FIG. 13 is a perspective view of another exemplary configuration of a multi-tap electrical connector.

[0023] FIG. 14 is a front view of FIG. 13.

[0024] FIG. 15 is an exploded view of FIG. 14.

[0025] FIG. 16 is a perspective view of another exemplary configuration of a multi-tap electrical connector.

[0026] FIG. 17 is a front view of FIG. 16.

[0027] FIG. 18 is an exploded view of FIG. 17.

[0028] FIG. 19 is a perspective view of another exemplary configuration of a multi-tap electrical connector.

[0029] FIG. 20 is a front view of FIG. 19.

[0030] FIG. 21 is an exploded view of FIG. 20.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0031] Tap connectors are used in different electrical distribution applications. Tap connectors are used to transfer electrical power between one or more conductors. For example, a primary, main, or run conductor and one or more secondary, branch, or tap conductors are physically and electrically connected, such as a distribution block, terminal lug, splice, or other connector. Power is transferred between the primary conductor and the tap conductors.

[0032] For example, tap conductors can be used in solar power generation facilities to help distribute power from individual solar panels to a main trunk line. Each panel or an array of panels can generate power which is fed to a junction box. Conductors are run from the junction box and can be connected to a combiner box. The combiner box combines the electrical outputs from multiple strings of panels into a single output. The main trunk line, also known as the bus bar, is a heavy-duty cable or set of cables that runs along the rows of solar panels. It collects the combined electrical output from all the parallel strings in the array. The trunk line typically carries the DC electricity to a central location, such as an inverter station or a substation, for further processing or distribution. The inverter converts the DC electricity from the solar panels into AC electricity suitable for grid connection or use in a local network.

[0033] As demand for solar power grows, installations generating solar power are trending toward larger footprints and larger production capacity. However, scaling solar farm capacity also grows complexity. Costs associated with installation, operation, and maintenance grow. Field terrain and other factors can challenge photovoltaic module layouts, requiring flexibility for connecting these installations. Accordingly, being able to produce flexible, repeatable, and reliable connections in a distribution system, such as a solar farm, offers a number of benefits.

[0034] Certain implementations are directed to connectors that are configured to connect a run line to multiple tap lines. Certain configurations can utilize insulation piercing teeth in one or both of the run lines and tap lines so that the need to strip conductors prior to installation is reduced or eliminated. Compression connectors with insulation piercing teeth are described in U.S. application Ser. No. 18/211,102, published as US 2023/0411871, the disclosure of which is hereby incorporated by reference in its entirety.

[0035] Run conductors are typically greater in size than the tap conductors. The run conductors and the tap conductors can be solid conductors, or they can be stranded conductors. Typically, the run conductors and tap conductors are stranded conductors. Different sizes and types of conductors can be used. Further, while the terms run/main/trunk conductors and secondary/tap/branch conductors may be used, the connectors described are suitable for mechanically and electrically joining any one conductor to any one or more other conductors regardless of the type of conductor or implementation of the power system.

[0036] The connections can be secured by regular bolt connectors or shear bolt connectors. The shear bolt can include a head connected to a threaded shaft by a shearable section. The shearable section can be a weakened area designed to break or otherwise separate once a certain torque limit is reached. As the fastener is rotated, the shaft or an associated pressure bar will engage a conductor. As the fastener is tightened, the engagement with the conductor will increase the amount of torque required to further turn the fastener. Once the torque limit is reached, the head will shear, separating from the fastener. This ensures a certain connection engagement is achieved with the conductor, while also preventing overtightening that could damage the conductor.

[0037] Use of shear bolt connectors can be accomplished with standard tooling, such as an electric impact driver. This can eliminate the need to use more complex tooling, such as a power crimping tool, which can be heavier and more expensive.

[0038] Use of these types of connectors can provide a number of advantages. In any electrical distribution system they can increase the flexibility of the system design and reduce the need for

complex tooling and installation procedures. They can also increase the repeatability and reliability of connections. In solar panel installations they can also eliminate the need for combiner boxes, allowing panels to be more directly connected to a trunk line.

[0039] FIGS. **1-4**, show an exemplary configuration of a multi-tap connector **100** having a conductor block **102**, a run line cap **104**, a run line bolt **106**, a plurality of tap line caps **108**, and a plurality of tap line bolts **110**. The illustrated configuration shows a single run line cap **104** and bolt **106** and four tap line caps **108** each with two tap line bolts **110**. Other configurations can utilize more or fewer run and tap lines as desired.

[0040] The conductor block **102** includes a body **112** having a substantially U-shaped configuration. The body **112** has a bottom portion **114**, a first side portion **116**, and a second side portion **118**. A run line opening **120** is formed in the body **112** between the first side portion **116** and the second side portion **118**. The run line opening **120** is configured to receive a run conductor. The size of the run line opening **120** can vary depending on the size or sizes of conductors that are to be received.

[0041] The run line opening **120** is bound on a first side by a first side wall **122** and on a second side by a second side wall **124**. A set of run insulation piercing members **126** extend into the run line opening **120**. The insulation piercing members **126** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **126** include a pair of outer teeth **128** and a central tooth **130**. The outer teeth **128** can extend higher than the central tooth **130** and can be angled toward the central tooth **130**.

[0042] Each of the teeth **128**, **130** can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **102**. The one or more insulation piercing members **126** can be integrally or monolithically formed into the bottom portion **114** of the conductor block **102** such that the one or more insulation piercing members **126** extend into the run opening **120**.

[0043] In other configurations, the one or more insulation piercing members **126** can be attachable to the body **112** of the conductor block **102**. For example, the insulation piercing members **126** can be formed as part of a removable assembly that can be mounted or secured to the body **112** of the conductor block **102**. Further, the insulation piercing members **126** can be made from a different material than the body **112**.

[0044] In certain implementations, performance of the insulation piercing members **126** can be improved by using a relatively harder material. The one or more insulation piercing members **126** can be made from a different material (e.g., copper, tempered aluminum, stainless steel, etc.) and secured to or attached to the conductor block **102** so that the one or more insulation piercing members **126** extend into the run opening **120**.

[0045] The insulation piercing members **126** are shown having a substantially triangular configuration. However, the insulation piercing members **126** can include different shapes and sizes configured and dimensioned to pierce or cut through the insulating jacket surrounding electrical wires of the run conductor, such as one or more cone-shaped members and/or one or more members with pointed tips. Further, one or more of the insulation piercing members **126** can include serrations on one or both sides of the insulation piercing members **126**.

[0046] In certain configurations, a pair of run guide slots **132** extend from the run line opening **120**. The run guide slots **132** are configured to slidably receive the run line cap **104**. In the illustrated configuration, the guide slots **132** include an inner wall **134**, an outer wall **136**, and a side wall **138**. The inner wall **134** extends further into the run opening **120** than the outer wall **136**. The outer wall **136** has an angled portion, forming a dovetail slot for receiving the run line cap **104** in the proper orientation.

[0047] Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run

cap **104** is integrally formed with the conductor block.

[0048] The run line cap **104** can include a body **140** configured to mate with the run guide slots **132** and fit into the conductor block **102**. In the illustrated configuration, the run line cap **104** includes an outer portion **142** and an inner portion **144**. The inner portion **144** is wider than the outer portion **142** to form a first shoulder **146** and a second shoulder **148**. An outer edge of the shoulders **146**, **148** is angled to correspond with the angled outer wall **136** of the run guide slots **132**.

[0049] The run line cap **104** slidably engages the conductor block **102** over the run line opening **120**. A run bolt opening **150** is provided in the run line cap **104** to receive the run line bolt **106**. The opening **150** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0050] In certain configurations the run line bolt **106** is a shear bolt having a shearable head **152**, a shank **154**, and a pressure bar **156**. The shank can include a thread and be threadably connected to the run line cap. The pressure bar **156** can be rotatably connected to the shank **154** and configured to engage the run conductor. In other configurations, the run line bolt **106** is a non-shear bolt.

[0051] The head **152** can be connected to the shank **154** by a shearable connection. The shearable section can be a weakened area designed to break or otherwise separate once a certain torque limit is reached. The shearable section can be formed by a frangible connection such as holes or slots placed in the material or can be formed from one or more section of reduced thickness. As the fastener is rotated, the shank **154** or an associated pressure bar **156** will engage a conductor. As the fastener is tightened, the engagement with the conductor will increase the amount of torque required to further turn the fastener. Once the torque limit is reached, the head **152** will shear, separating from the shank **154**. This ensures a certain connection engagement is achieved with the conductor, while also preventing overtightening that could damage the conductor. Different shear limits can be used according to different conductors.

[0052] Similar to the run line connection, the conductor block **102** can include multiple tap openings for allowing connections to be formed with tap conductors. In certain configurations, the first side portion **116** and the second side portion **118** of the conductor block **102** each include two sets of tap line openings **158**. The tap line openings **158** are bound on a first side by a first side wall **160** and on a second side by a second side wall **162**. A set of tap insulation piercing members **164** extend into each of the tap line openings **158**. The insulation piercing members **164** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **164** include a pair of outer teeth **166** and a central tooth **168**. The outer teeth **166** can extend farther than the central tooth **168** and can be angled toward the central tooth **168**. The tap insulation piercing members **164** can have any of the same properties as the run insulation piercing members **126** described herein, including being made from different materials, having different configurations, and being formed as a separate assembly releasably connected to the conductor block **102**.

[0053] A pair of tap guide slots **170** extend from each of the tap line openings **158**. The tap guide slots **170** can have a similar configuration as the run guide slots **132**. The tap guide slots **170** include an inner wall **172**, an outer wall **174**, and a side wall **176**. The inner wall **172** extends further into the tap opening **158** than the outer wall **174**. The outer wall **174** has an angled portion, forming a dovetail slot for receiving the tap line cap **108** in the proper orientation.

[0054] Other configurations can utilize different sizes and shapes of tap guide slots **170**, including squared and rounded configurations. Fewer or more tap guide slots **170** can also be incorporated into the conductor block **102**. Certain configurations can also utilize a unitary body where the tap line caps **108** are integrally formed with the conductor block **102**.

[0055] The tap line caps **108** can include a body **178** configured to mate with the tap guide slots **170** and fit into the conductor block **102**. In the illustrated configuration, the tap line caps **108** include an outer portion **180** and an inner portion **182**. The inner portion **182** is wider than the outer portion **180** to form a first shoulder **184** and a second shoulder **186**. An outer edge of the shoulders **184**, **186** is angled to correspond with the angled outer wall **174** of the tap guide slots **170**. The tap

line caps **108** slidably engages the conductor block **102** over the tap line openings **158**. [0056] In the illustrated configuration, two bolt openings **188** are provided in each of the tap line caps **108** to receive a pair of tap line bolts **110**. The openings **188** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **188** can be used as needed. The use of two tap bolt openings **188** can allow each tap line opening **158** to receive two tap lines, one from each side. Certain configurations can utilize an internal divider, as part of the conductor block **102** or tap line caps **108**, as needed, to separate each tap line opening **158** into a first and second chamber. This can help a user ensure proper seating of the tap line conductors.

[0057] Each tap line bolt **110** can include a head **190**, a shank **192**, and a pressure bar **194**. The shank **192** can include a thread and be threadably connected to the tap line cap **108**. The pressure bar **194** can be rotatably connected to the shank **192** and configured to engage the run conductor. The head **190** can be connected to the shank **192** by a shearable connection as discussed with respect to the run line shear bolt **106**. In other configurations, the head is a non-shear bolt.

[0058] During installation, a tap conductor can be placed into a tap opening **158**. The associated tap bolt **110** is then tightened by rotating the head **190** so that the tap pressure bar **194** engages the tap conductor. The tap bolt **110** is then tightened further so that the tap conductor is driven into the tap insulation piercing members **164**. The insulation piercing members **164** will pierce the outer insulation of the tap conductor to form an electrical and mechanical connection with the tap conductor and the conductor block **102**. At a certain point, the shear limit of the head **190** will be reached, resulting in the head **190** separating from the shank **192** and preventing overtightening. Additional tap conductors can be connected in this manner as needed. Similarly, the run conductor can be placed in the run line opening **120**. The run bolt **106** is then tightened by rotating the head **152** so that the tap pressure bar **156** engages the run conductor. The run bolt **106** is then tightened further so that the run conductor is driven into the run insulation piercing members **126**. The insulation piercing members **126** will pierce the outer insulation of the tap conductor to form an electrical and mechanical connection with the tap conductor and the conductor block **102**. At a certain point, the shear limit of the head **152** will be reached, resulting in the head **152** separating from the shank **154** and preventing overtightening. The run and tap conductors can be connected to the conductor block **102** in this manner in any order.

[0059] The use of the shear bolts **106**, **110** and the insulation piercing members **126**, **164** simplifies the installation procedure and can result in a faster more repeatable connection. For example, the need to strip conductors before connection to the conductor block **102** can be eliminated and lighter tools, such as an impact driver, can be used instead of heavier crimping tools. However, advantages of easier and more reliable connections are also achieved without the use of shear bolts.

[0060] In some configurations, the conductor block **102** and body **112** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **102**. For example, the body **112** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the bottom of the block to add additional tap openings **158** to connect off of the existing run opening **120**.

[0061] FIGS. 5 and 6, show an exemplary configuration of a multi-tap connector **200** having a conductor block **202**, a run line cap **204**, a run line bolt **206**, a plurality of tap line caps **208**, and a plurality of tap line bolts **210**. The illustrated configuration shows a single run line cap **204** and bolt **206** and two tap line caps **208** each with two tap line bolts **210**. Other configurations can utilize more or fewer run and tap lines as desired.

[0062] The conductor block **202** includes a body **212** having a substantially L-shaped configuration. The body **212** has a run side portion **216** and a tap side portion **218**. A run line opening **220** is formed in run side portion **216**. The run line opening **220** is configured to receive a run conductor. The size of the run line opening **220** can vary depending on the size or sizes of

conductors that are to be received.

[0063] The run line opening **220** is bound on a first side by a first side wall **222** and on a second side by a second side wall **224**. A set of run insulation piercing members **226** extend into the run line opening **220**. The insulation piercing members **226** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **226** include a pair of outer teeth **228** and a central tooth **230**. The outer teeth **228** can extend higher than the central tooth **230** and can be angled toward the central tooth **230**.

[0064] Each of the teeth **228**, **230** can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **202**. The one or more insulation piercing members **226** can be integrally or monolithically formed into the bottom portion **214** of the conductor block **202** such that the one or more insulation piercing members **226** extend into the run opening **220**. In other configurations, the one or more insulation piercing members **226** can be attachable to the body **212** of the conductor block **202**.

[0065] For example, the insulation piercing members **226** can be formed as part of a removable assembly that can be mounted or secured to the body **212** of the conductor block **202**. Further, the insulation piercing members **226** can be made from a different material than the body **212**.

[0066] In certain configurations, a pair of run guide slots **232** extend from the run line opening **220**. The run guide slots **232** are configured to slidably receive the run line cap **204**. In the illustrated configuration, the guide slots **232** include an inner wall **234**, an outer wall **236**, and a side wall **238**. The inner wall **234** extends further into the run opening **220** than the outer wall **236**. The outer wall **236** has an angled portion, forming a dovetail slot for receiving the run line cap **204** in the proper orientation.

[0067] Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run cap **204** is integrally formed with the conductor block.

[0068] The run line cap **204** can include a body **240** configured to mate with the run guide slots **232** and fit into the conductor block **202**. In the illustrated configuration, the run line cap **204** includes an outer portion **242** and an inner portion **244**. The inner portion **244** is wider than the outer portion **242** to form a first shoulder **246** and a second shoulder **248**. An outer edge of the shoulders **246**, **248** is angled to correspond with the angled outer wall **236** of the run guide slots **232**.

[0069] The run line cap **204** slidably engages the conductor block **202** over the run line opening **220**. A run bolt opening **250** is provided in the run line cap **204** to receive the run line bolt **206**. The opening **250** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0070] In certain configurations the run line bolt **206** is a shear bolt having a shearable head **252**, a shank **254**, and a pressure bar **256**. The shank can include a thread and be threadably connected to the run line cap **204**. The pressure bar **256** can be rotatably connected to the shank **254** and configured to engage the run conductor. In other configurations, the run line bolt **206** is a non-shear bolt.

[0071] The head **252** can be connected to the shank **254** by a shearable connection having a torque limit. As the fastener is rotated, the shank **254** or an associated pressure bar **256** will engage a conductor. As the fastener is tightened, the engagement with the conductor will increase the amount of torque required to further turn the fastener. Once the torque limit is reached, the head **252** will shear, separating from the shank **254**. This ensures a certain connection engagement is achieved with the conductor, while also preventing overtightening that could damage the conductor. Different shear limits can be used according to different conductors.

[0072] Similar to the run line connection, the conductor block **202** can include multiple tap openings for allowing connections to be formed with tap conductors. In certain configurations, the tap side portion **228** of the conductor block **202** includes two sets of tap line openings **258**. The tap line openings **258** are bound on a first side by a first side wall **260** and on a second side by a second

side wall **262**. A set of tap insulation piercing members **264**, extend into each of the tap line openings **258**. The insulation piercing members **264** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **264** include a pair of outer teeth **266** and a central tooth **268**. The outer teeth **266** can extend farther than the central tooth **268** and can be angled toward the central tooth **268**. The tap insulation piercing members **264** can have any of the same properties as the run insulation piercing members **226** described herein, including being made from different materials, having different configurations, and being formed as a separate assembly releasably connected to the conductor block **202**.

[0073] A pair of tap guide slots **270** extend from each of the tap line openings **258**. The tap guide slots **270** can have a similar configuration as the run guide slots **232**. The tap guide slots **270** include an inner wall **272**, an outer wall **274**, and a side wall **276**. The inner wall **272** extends further into the tap opening **258** than the outer wall **274**. The outer wall **274** has an angled portion, forming a dovetail slot for receiving the tap line cap **208** in the proper orientation.

[0074] Other configurations can utilize different sizes and shapes of tap guide slots **270**, including squared and rounded configurations. Fewer or more tap guide slots **270** can also be incorporated into the conductor block **202**. Certain configurations can also utilize a unitary body where the tap line caps **208** are integrally formed with the conductor block **202**.

[0075] The tap line caps **208** can include a body **278** configured to mate with the tap guide slots **270** and fit into the conductor block **202**. In the illustrated configuration, the tap line caps **208** include an outer portion **280** and an inner portion **282**. The inner portion **282** is wider than the outer portion **280** to form a first shoulder **284** and a second shoulder **286**. An outer edge of the shoulders **284**, **286** is angled to correspond with the angled outer wall **274** of the tap guide slots **270**. The tap line caps **208** slidably engages the conductor block **202** over the tap line openings **258**.

[0076] In the illustrated configuration, two bolt openings **288** are provided in each of the tap line caps **208** to receive a pair of tap line bolts **210**. The openings **288** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **288** can be used as needed. The use of two tap bolt openings **288** can allow each tap line opening **258** to receive two tap lines, one from each side. Certain configurations can utilize an internal divider, as part of the conductor block **202** or tap line caps **208**, as needed, to separate each tap line opening **258** into a first and second chamber. This can help a user ensure proper seating of the tap line conductors.

[0077] Each tap line bolt **210** can include a head **290**, a shank **292**, and a pressure bar **294**. The shank **292** can include a thread and be threadably connected to the tap line cap **208**. The pressure bar **294** can be rotatably connected to the shank **292** and configured to engage the run conductor. The head **290** can be connected to the shank **292** by a shearable connection as discussed with respect to the run line shear bolt **206**. In other configurations, the head **290** is a non-shear bolt.

[0078] In some configurations, the conductor block **202** body **212** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **202**. For example, the body **212** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the tap side portion **218** of the block to add additional tap openings **258** to connect off of the existing run opening **220**. The tap side portion **218** can also be formed to include additional tap line openings **258** as required.

[0079] FIGS. 7-9, show an exemplary configuration of a multi-tap connector **300** having a conductor block **302**, a run line cap **304**, a run line bolt **306**, a plurality of tap line caps **308**, and a plurality of tap line bolts **310**. The illustrated configuration shows a single run line cap **304** and bolt **306** and two tap line caps **308** each with two tap line bolts **310**. Other configurations can utilize more or fewer run and tap lines as desired.

[0080] The conductor block **302** includes a body **312** having a run side portion **316** and a tap side

portion **318**. A run line opening **320** is formed in the run side portion **316**. The run line opening **320** is configured to receive a run conductor. The size of the run line opening **320** can vary depending on the size or sizes of conductors that are to be received.

[0081] The run line opening **320** is bound on a first side by a first side wall **322** and on a second side by a second side wall **324**. A set of run insulation piercing members **326** extend into the run line opening **320**. The insulation piercing members **326** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **326** include a pair of outer teeth **328** and a central tooth **330**. The outer teeth **328** can extend higher than the central tooth **330** and can be angled toward the central tooth **330**.

[0082] Each of the teeth **328**, **330** can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **302**. The one or more insulation piercing members **326** can be integrally or monolithically formed into the conductor block **302** such that the one or more insulation piercing members **326** extend into the run opening **320**. In other configurations, the one or more insulation piercing members **326** can be attachable to the body **312** of the conductor block **302**.

[0083] For example, the insulation piercing members **326** can be formed as part of a removable assembly that can be mounted or secured to the body **312** of the conductor block **302**. Further, the insulation piercing members **326** can be made from a different material than the body **312**.

[0084] In certain configurations, a pair of run guide slots **332** extend from the run line opening **320**. The run guide slots **332** are configured to slidably receive the run line cap **304**. In the illustrated configuration, the guide slots **332** include an inner wall **334**, an outer wall **336**, and a side wall **338**. The inner wall **334** extends further into the run opening **220** than the outer wall **336**. The outer wall **336** has an angled portion, forming a dovetail slot for receiving the run line cap **304** in the proper orientation.

[0085] Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run cap **304** is integrally formed with the conductor block.

[0086] The run line cap **304** can include a body **340** configured to mate with the run guide slots **332** and fit into the conductor block **302**. In the illustrated configuration, the run line cap **304** includes an outer portion **342** and an inner portion **344**. The inner portion **344** is wider than the outer portion **342** to form a first shoulder **346** and a second shoulder **348**. An outer edge of the shoulders **346**, **348** is angled to correspond with the angled outer wall **336** of the run guide slots **332**.

[0087] The run line cap **304** slidably engages the conductor block **302** over the run line opening **320**. A run bolt opening **350** is provided in the run line cap **304** to receive the run line bolt **306**. The opening **350** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0088] In certain configurations the run line bolt **306** is a shear bolt having a shearable head **352**, a shank **354**, and a pressure bar **356**. The shank can include a thread and be threadably connected to the run line cap **304**. The pressure bar **356** can be rotatably connected to the shank **354** and configured to engage the run conductor. In other configurations, the run line bolt **306** is a non-shear bolt.

[0089] The head **352** can be connected to the shank **354** by a shearable connection having a torque limit. As the fastener is rotated, the shank **354** or an associated pressure bar **356** will engage a conductor. As the fastener is tightened, the engagement with the conductor will increase the amount of torque required to further turn the fastener. Once the torque limit is reached, the head **352** will shear, separating from the shank **354**. This ensures a certain connection engagement is achieved with the conductor, while also preventing overtightening that could damage the conductor. Different shear limits can be used according to different conductors.

[0090] Similar to the run line connection, the conductor block **302** can include multiple tap openings for allowing connections to be formed with tap conductors. In certain configurations, the

tap side portion **328** of the conductor block **302** each include two sets of tap line openings **358**. The tap line openings **358** are bound on a first side by a first side wall **360** and on a second side by a second side wall **362**. A set of tap insulation piercing members **364**, extend into each of the tap line openings **358**. The insulation piercing members **364** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **364** include a pair of outer teeth **366** and a central tooth **368**. The outer teeth **366** can extend farther than the central tooth **368** and can be angled toward the central tooth **368**. The tap insulation piercing members **364** can have any of the same properties as the run insulation piercing members **326** described herein, including being made from different materials, having different configurations, and being formed as a separate assembly releasably connected to the conductor block **302**.

[0091] A pair of tap guide slots **370** extend from each of the tap line openings **358**. The tap guide slots **370** can have a similar configuration as the run guide slots **332**. The tap guide slots **370** include an inner wall **372**, an outer wall **374**, and a side wall **376**. The inner wall **372** extends further into the tap opening **358** than the outer wall **374**. The outer wall **374** has an angled portion, forming a dovetail slot for receiving the tap line cap **308** in the proper orientation.

[0092] Other configurations can utilize different sizes and shapes of tap guide slots **370**, including squared and rounded configurations. Fewer or more tap guide slots **370** can also be incorporated into the conductor block **302**. Certain configurations can also utilize a unitary body where the tap line caps **308** are integrally formed with the conductor block **302**.

[0093] The tap line caps **308** can include a body **378** configured to mate with the tap guide slots **370** and fit into the conductor block **302**. In the illustrated configuration, the tap line caps **308** include an outer portion **380** and an inner portion **382**. The inner portion **382** is wider than the outer portion **380** to form a first shoulder **384** and a second shoulder **386**. An outer edge of the shoulders **384**, **386** is angled to correspond with the angled outer wall **374** of the tap guide slots **370**. The tap line caps **308** slidably engages the conductor block **302** over the tap line openings **358**.

[0094] In the illustrated configuration, two bolt openings **388** are provided in each of the tap line caps **308** to receive a pair of tap line bolts **310**. The openings **388** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **388** can be used as needed. The use of two tap bolt openings **388** can allow each tap line opening **358** to receive two tap lines, one from each side. Certain configurations can utilize an internal divider, as part of the conductor block **302** or tap line caps **308**, as needed, to separate each tap line opening **358** into a first and second chamber. This can help a user ensure proper seating of the tap line conductors.

[0095] Each tap line bolt **310** can include a head **390**, a shank **392**, and a pressure bar **394**. The shank **392** can include a thread and be threadably connected to the tap line cap **308**. The pressure bar **394** can be rotatably connected to the shank **392** and configured to engage the run conductor. The head **390** can be connected to the shank **392** by a shearable connection as discussed with respect to the run line shear bolt **306**. In other configurations, the head **390** is a non-shear bolt.

[0096] In some configurations, the conductor block **302** body **312** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **302**. For example, the body **312** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the tap sides of the block **302** to add additional tap openings **358** to connect off of the existing run opening **320**.

[0097] FIGS. **10-12** show alternative embodiments of the conductor blocks **102**, **202**, **302** having run bolts and tap bolts without pressure bars. FIG. **10** shows a connector **400** having a conductor block **402** and a run bolt **404** and a set of tap bolts **410**. The run bolt **404** has a convex end **412** for engaging the run conductor and the tap bolts **410** have a convex end **414** for engaging the tap conductors. FIG. **11** shows a connector **500** having a conductor block **502** and a run bolt **504** and a

set of tap bolts **510**. The run bolt **504** has a convex end **512** for engaging the run conductor and the tap bolts **510** have a convex end **514** for engaging the tap conductors. FIG. **12** shows a connector **600** having a conductor block **602** and a run bolt **604** and a set of tap bolts **610**. The run bolt **604** has a convex end **612** for engaging the run conductor and the tap bolts **610** have a convex end **614** for engaging the tap conductors.

[0098] Although the exemplary configurations described utilize a shear bolt with a single head, different kinds of shear bolts can be used. For example, shear bolts having more than one head can be used to set different shear limits or to provide a non-shear head to allow for easier removal of the fastener after the initial head is sheared. Different pressure bars, pressure discs, and no pressure bar configurations can also be used. Further configurations can implement insulation piercing as part of the fastener instead of, or in addition to, the insulation piercing members associated with the conductor blocks. Other embodiments can utilize non-shear bolts, with torque limits eliminated or set by other ways on the fastener or as part of the installation tool.

[0099] FIGS. **13-15**, show an exemplary configuration of a multi-tap connector **700** having a conductor block **702**, a run line cap **704**, a run line bolt **706**, a plurality of run line openings **708** extending into the conductor block **702**, and a plurality of tap line bolts **710**. The illustrated configuration shows a single run line cap **704** and bolt **706** and four tap line openings **708** each with two tap line bolts **710**. Other configurations can utilize more or fewer run and tap lines as desired.

[0100] The conductor block **702** includes a body **712** having a substantially U-shaped configuration. The body **712** has a bottom portion **714**, a first side portion **716**, and a second side portion **718**. A run line opening **720** is formed in the body **712** between the first side portion **716** and the second side portion **718**. The run line opening **720** is configured to receive a run conductor. The size of the run line opening **720** can vary depending on the size or sizes of conductors that are to be received.

[0101] The run line opening **720** is bound on a first side by a first side wall and on a second side by a second side wall. A set of run insulation piercing members **726** extend into the run line opening **720**. The insulation piercing members **726** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **726** include a plurality of teeth extending in an arc.

[0102] Each of the teeth can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **702**. The one or more insulation piercing members **726** can be integrally or monolithically formed into the bottom portion **714** of the conductor block **702** such that the one or more insulation piercing members **726** extend into the run opening **720**.

[0103] In other configurations, the one or more insulation piercing members **726** can be attachable to the body **712** of the conductor block **702**. For example, the insulation piercing members **726** can be formed as part of a removable assembly that can be mounted or secured to the body **712** of the conductor block **702**. Further, the insulation piercing members **726** can be made from a different material than the body **712**.

[0104] In certain implementations, performance of the insulation piercing members **726** can be improved by using a relatively harder material. The one or more insulation piercing members **726** can be made from a different material (e.g., copper, tempered aluminum, stainless steel, etc.) and secured to or attached to the conductor block **702** so that the one or more insulation piercing members **726** extend into the run opening **720**.

[0105] The insulation piercing members **726** are shown having a substantially triangular configuration. However, the insulation piercing members **726** can include different shapes and sizes configured and dimensioned to pierce or cut through the insulating jacket surrounding electrical wires of the run conductor, such as one or more cone-shaped members and/or one or more

members with pointed tips. Further, one or more of the insulation piercing members **726** can include serrations on one or both sides of the insulation piercing members **726**.

[0106] In certain configurations, a pair of run guide slots **732** extend from the run line opening **720**. The run guide slots **732** are configured to slidably receive the run line cap **704**. In the illustrated configuration, the guide slots **732** include an inner wall **734**, an outer wall **736**, and a side wall **738**. The inner wall **734** extends further into the run opening **720** than the outer wall **736**.

[0107] Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run cap **704** is integrally formed with the conductor block.

[0108] The run line cap **704** can include a body **740** configured to mate with the run guide slots **732** and fit into the conductor block **702**. In the illustrated configuration, the run line cap **704** includes an outer portion **742** and an inner portion **744**. The inner portion **744** is wider than the outer portion **742** to form a first shoulder **746** and a second shoulder **748**. The shoulders **746**, **748** correspond with the run guide slots **732**.

[0109] The run line cap **704** slidably engages the conductor block **702** over the run line opening **720**. A run bolt opening **750** is provided in the run line cap **704** to receive the run line bolt **706**. The opening **750** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0110] In certain configurations the run line bolt **706** is a shear bolt having a shearable head **752**, a shank **754**, and a pressure bar **756**. The shank can include a thread and be threadably connected to the run line cap. The pressure bar **756** can be rotatably connected to the shank **754** and configured to engage the run conductor. In other configurations, the run line bolt **706** is a non-shear bolt.

[0111] In certain configurations, the pressure bar **756** can include a set of insulation piercing members **758** that extend from the pressure bar **756** toward the insulation piercing members **726** of the run line opening **720**. In the illustrated configuration, the pressure bar insulation piercing members **758** include a plurality of teeth extending in an arc. Other configurations can also be used.

[0112] The conductor block **702** can include multiple tap openings **708** for allowing connections to be formed with tap conductors. In certain configurations, the first side portion **716** and the second side portion **718** of the conductor block **702** each include two sets of tap line openings **708**. A set of tap insulation piercing members **764**, extend into each of the tap line openings **708**. The insulation piercing members **764** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor.

[0113] In the illustrated configuration, two bolt openings **788** are provided in communication with each of the tap line openings **708** to receive a pair of tap line bolts **710**. The bolt openings **788** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **788** can be used as needed. The use of two tap bolt openings **788** can allow each tap line opening **708** to receive two tap lines, one from each side. Certain configurations can utilize an internal divider, as part of the conductor block **702**, as needed, to separate each tap line opening **708** into a first and second chamber. This can help a user ensure proper seating of the tap line conductors.

[0114] Each tap line bolt **710** can include a head **790** and a shank **792**. The shank **792** can include a thread and be threadably connected to the bolt opening **788**. The head **790** can be connected to the shank **792** by a shearable connection as discussed with respect to the run line shear bolt **706**. In other configurations, the head **790** is a non-shear bolt.

[0115] In some configurations, the conductor block **702** body **712** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **702**. For example, the body **712** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the bottom of the block to add additional tap openings **708** to connect off of the existing run opening **720**.

[0116] FIGS. **16-18**, show an exemplary configuration of a multi-tap connector **800** having a

conductor block **802**, a run line cap **804**, a run line bolt **806**, a plurality of tap line openings **808**, and a plurality of tap line bolts **810**. The illustrated configuration shows a single run line cap **804** and bolt **806** and two tap line openings **808** each with a single tap line bolt **810**. Other configurations can utilize more or fewer run and tap lines as desired.

[0117] The conductor block **802** includes a body **812** having a run side portion **816** and a tap side portion **818**. A run line opening **820** is formed in run side portion **816**. The run line opening **820** is configured to receive a run conductor. The size of the run line opening **820** can vary depending on the size or sizes of conductors that are to be received.

[0118] The run line opening **820** is bound on a first side by a first side wall and on a second side by a second side wall. A set of run insulation piercing members **826** extend into the run line opening **820**. The insulation piercing members **826** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **826** include a plurality of teeth extending in an arc.

[0119] The teeth can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **802**. The one or more insulation piercing members **826** can be integrally or monolithically formed into the conductor block **802** such that the one or more insulation piercing members **826** extend into the run opening **820**. In other configurations, the one or more insulation piercing members **826** can be attachable to the body **812** of the conductor block **802**.

[0120] For example, the insulation piercing members **826** can be formed as part of a removable assembly that can be mounted or secured to the body **812** of the conductor block **802**. Further, the insulation piercing members **826** can be made from a different material than the body **812**.

[0121] In certain configurations, a pair of run guide slots **832** extend from the run line opening **820**. The run guide slots **832** are configured to slidably receive the run line cap **804**. The run guide slots have a rectangular configuration with an inner wall extending further than an outer wall. Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run cap **804** is integrally formed with the conductor block.

[0122] The run line cap **804** can include a body **840** configured to mate with the run guide slots **832** and fit into the conductor block **802**. In the illustrated configuration, the run line cap **804** includes an outer portion **842** and an inner portion **844**. The inner portion **844** is wider than the outer portion **842** to form a first shoulder **846** and a second shoulder **848**. The run line cap **804** slidably engages the conductor block **802** over the run line opening **820**. A run bolt opening **850** is provided in the run line cap **804** to receive the run line bolt **806**. The opening **850** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0123] In certain configurations the run line bolt **806** is a shear bolt having a shearable head **852**, a shank **854**, and a pressure bar **856**. The shank can include a thread and be threadably connected to the run line cap **804**. The pressure bar **856** can be rotatably connected to the shank **854** and configured to engage the run conductor. In other configurations, the run line bolt **806** is a non-shear bolt.

[0124] In certain configurations, the pressure bar **856** can include a set of insulation piercing members **858** that extend from the pressure bar **856** toward the insulation piercing members **826** of the run line opening **820**. In the illustrated configuration, the pressure bar insulation piercing members **858** include a plurality of teeth extending in an arc. Other configurations can also be used.

[0125] The head **852** can be connected to the shank **854** by a shearable connection having a torque limit. As the fastener is rotated, the shank **854** or an associated pressure bar **856** will engage a conductor. As the fastener is tightened, the engagement with the conductor will increase the amount of torque required to further turn the fastener. Once the torque limit is reached, the head **852** will shear, separating from the shank **854**. This ensures a certain connection engagement is achieved

with the conductor, while also preventing overtightening that could damage the conductor.

Different shear limits can be used according to different conductors. In other configurations, the head **850** is a non-shear bolt.

[0126] Similar to the run line connection, the conductor block **802** can include multiple tap openings **808** for allowing connections to be formed with tap conductors. In certain configurations, the tap side portion **828** of the conductor block **802** includes two tap line openings **808**. The tap line openings **808** can be open to the exterior of the conductor block **802**, as shown, or closed. A set of tap insulation piercing members **864** extend into each of the tap line openings **808**. The insulation piercing members **864** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor.

[0127] In the illustrated configuration, a bolt opening **888** is provided in each of the tap line openings **808** to receive a pair of tap line bolts **810**. The openings **888** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **888** can be used as needed.

[0128] Each tap line bolt **810** can include a head **890** and a shank **892**. The shank **892** can include a thread and be threadably connected to the tap line cap **808**. The head **890** can be connected to the shank **892** by a shearable connection as discussed with respect to the run line shear bolt **806**. In other configurations, the head **890** is a non-shear bolt.

[0129] In some configurations, the conductor block **802** body **812** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **802**. For example, the body **812** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the tap sides of the block **802** to add additional tap openings **808** to connect off of the existing run opening **820**.

[0130] FIGS. **19-21**, show an exemplary configuration of a multi-tap connector **900** having a conductor block **902**, a run line cap **904**, a run line bolt **906**, a plurality of run line openings **908** extending into the conductor block **902**, and a plurality of tap line bolts **910**. The illustrated configuration shows a single run line cap **904** and bolt **906** and four tap line openings **908** each with two tap line bolts **910**. Other configurations can utilize more or fewer run and tap lines as desired.

[0131] The conductor block **902** includes a body **912** having a substantially rounded U-shaped configuration with curved sides. The body **912** has a bottom portion **914**, a first side portion **916**, and a second side portion **918**. A run line opening **920** is formed in the body **912** between the first side portion **916** and the second side portion **918**. The run line opening **920** is configured to receive a run conductor. The size of the run line opening **920** can vary depending on the size or sizes of conductors that are to be received.

[0132] The run line opening **920** is bound on a first side by a first side wall and on a second side by a second side wall. A set of run insulation piercing members **926** extend into the run line opening **920**. The insulation piercing members **926** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor. In certain configurations, the insulation piercing members **926** include a plurality of teeth extending in an arc.

[0133] Each of the teeth can include an insulation piercing tip that is configured and dimensioned to pierce or cut through the insulating jacket of the run conductor to create an electrical path between the run conductor and the conductor block **902**. The one or more insulation piercing members **926** can be integrally or monolithically formed into the bottom portion **914** of the conductor block **902** such that the one or more insulation piercing members **926** extend into the run opening **920**.

[0134] In other configurations, the one or more insulation piercing members **926** can be attachable to the body **912** of the conductor block **902**. For example, the insulation piercing members **926** can be formed as part of a removable assembly that can be mounted or secured to the body **912** of the

conductor block **902**. Further, the insulation piercing members **926** can be made from a different material than the body **912**.

[0135] In certain implementations, performance of the insulation piercing members **926** can be improved by using a relatively harder material. The one or more insulation piercing members **926** can be made from a different material (e.g., copper, tempered aluminum, stainless steel, etc.) and secured to or attached to the conductor block **902** so that the one or more insulation piercing members **926** extend into the run opening **920**.

[0136] The insulation piercing members **926** are shown having a substantially triangular configuration. However, the insulation piercing members **926** can include different shapes and sizes configured and dimensioned to pierce or cut through the insulating jacket surrounding electrical wires of the run conductor, such as one or more cone-shaped members and/or one or more members with pointed tips. Further, one or more of the insulation piercing members **926** can include serrations on one or both sides of the insulation piercing members **926**.

[0137] In certain configurations, a pair of run guide slots **932** extend from the run line opening **920**. The run guide slots **932** are configured to slidably receive the run line cap **904**. In the illustrated configuration, the guide slots **932** include an inner wall, an outer wall, and a side wall. The inner wall extends further into the run opening **920** than the outer wall.

[0138] Other configurations can utilize different sizes and shapes of guide slots, including squared and rounded configurations. Certain configurations can also utilize a unitary body where the run cap **904** is integrally formed with the conductor block.

[0139] The run line cap **904** can include a body **940** configured to mate with the run guide slots **932** and fit into the conductor block **902**. In the illustrated configuration, the run line cap **904** includes an outer portion **942** and an inner portion **944**. The outer portion **942** can have a curved upper edge corresponding to the curved sides of the conductor block body **912**. The inner portion **944** is wider than the outer portion **942** to form a first shoulder **946** and a second shoulder **948**. The shoulders **946**, **948** correspond with the run guide slots **932**.

[0140] The run line cap **904** slidably engages the conductor block **902** over the run line opening **920**. A run bolt opening **950** is provided in the run line cap **904** to receive the run line bolt **906**. The opening **950** can be threaded and sized to accommodate the bolt appropriate for the run conductor.

[0141] In certain configurations the run line bolt **906** is a shear bolt having a shearable head **952**, a shank **954**, and a pressure bar **956**. The shank can include a thread and be threadably connected to the run line cap. The pressure bar **956** can be rotatably connected to the shank **954** and configured to engage the run conductor. In other configurations, the run line bolt **906** is a non-shear bolt.

[0142] In certain configurations, the pressure bar **956** can include a set of insulation piercing members **958** that extend from the pressure bar **956** toward the insulation piercing members **926** of the run line opening **920**. In the illustrated configuration, the pressure bar insulation piercing members **958** include a plurality of teeth extending in an arc. Other configurations can also be used.

[0143] The conductor block **902** can include multiple tap openings **908** for allowing connections to be formed with tap conductors. In certain configurations, the first side portion **916** and the second side portion **918** of the conductor block **902** each include two sets of tap line openings **908**. A set of tap insulation piercing members **964**, extend into each of the tap line openings **908**. The insulation piercing members **964** can be teeth, tines, or other features configured to pierce the insulating jacket of a conductor.

[0144] In the illustrated configuration, two bolt openings **988** are provided in communication with each of the tap line openings **908** to receive a pair of tap line bolts **910**. The bolt openings **988** can be threaded and sized to accommodate the bolt appropriate for the tap conductor. Fewer or more tap bolt openings **988** can be used as needed. The use of two tap bolt openings **988** can allow each tap line opening **908** to receive two tap lines, one from each side. Certain configurations can utilize an internal divider, as part of the conductor block **902**, as needed, to separate each tap line opening **908** into a first and second chamber. This can help a user ensure proper seating of the tap line

conductors.

[0145] Each tap line bolt **910** can include a head **990** and a shank **992**. The shank **992** can include a thread and be threadably connected to the bolt opening **988**. The head **990** can be connected to the shank **992** by a shearable connection as discussed with respect to the run line shear bolt **906**. In other configurations, the head **990** is a non-shear bolt.

[0146] In some configurations, the conductor block **902** body **912** can be modified to connect to other conductor blocks or block extensions to increase the number of tap lines that can be connected to the block **902**. For example, the body **912** can include mating features, such as male and female mating features that can allow other blocks or extensions to be added on in the field as needed. These could be connected to the bottom of the block to add additional tap openings **908** to connect off of the existing run opening **920**.

[0147] The foregoing detailed description of the certain exemplary configurations and implementations has been provided for the purpose of explaining the general principles and practical application, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with various modifications as are suited to the particular use contemplated. This description is not necessarily intended to be exhaustive or to limit the disclosure to the exemplary embodiments disclosed. Any of the embodiments and/or elements disclosed herein may be combined with one another to form various additional embodiments not specifically disclosed. Accordingly, additional embodiments are possible and are intended to be encompassed within this specification and the scope of the appended claims. The specification describes specific examples to accomplish a more general goal that may be accomplished in another way.

[0148] As used in this application, the terms “front,” “rear,” “upper,” “lower,” “upwardly,” “downwardly,” and other orientational descriptors are intended to facilitate the description of the exemplary embodiments of the present disclosure, and are not intended to limit the structure of the exemplary embodiments of the present disclosure to any particular position or orientation. Terms of degree, such as “substantially” or **41** “approximately” are understood by those of ordinary skill to refer to reasonable ranges outside of the given value, for example, general tolerances associated with manufacturing, assembly, and use of the described embodiments. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0149] Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

Claims

1. An electrical connector for connecting a plurality of conductors comprising: a conductor block having a body with a run line opening, a run insulation piercing member extending into the run line opening, a tap line opening, and a tap insulation piercing member extending into the tap line opening; a run cap extending over the run line opening; a run bolt extending through the run cap into the run line opening, wherein the run bolt includes a first threaded shank and a first shearable head, wherein the first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit; a tap cap extending over the tap line opening; and a tap bolt extending through the tap cap into the tap line opening, wherein the tap bolt includes a second threaded shank and a second shearable head, wherein the second shearable head is configured to disconnect from

the second threaded shank on reaching a second torque limit.

2. The electrical connector of claim 1, wherein the run insulation piercing member is integrally formed with the conductor block.

3. The electrical connector of claim 1, wherein the run cap is removably connected to the conductor block.

4. The electrical connector of claim 1, wherein the run cap is slidably connected to the conductor block.

5. The electrical connector of claim 1, wherein the tap cap is removably connected to the conductor block.

6. The electrical connector of claim 1, wherein the run insulation piercing member includes a set of teeth.

7. The electrical connector of claim 1, wherein the run bolt includes a pressure bar connected to the first shank.

8. The electrical connector of claim 1, wherein the tap opening is a first tap opening, the conductor block further comprises a second tap opening, and the run opening is positioned between the first tap opening and the second tap opening.

9. The electrical connector of claim 1, wherein the conductor block has a substantially L-shaped configuration with the body having a run portion and a tap portion extending from the run portion.

10. The electrical connector of claim 1, wherein the tap cap includes a first tap bolt opening receiving the tap bolt and a second tap bolt opening receiving a second tap bolt.

11. An electrical connector for connecting a plurality of conductors comprising: a conductor block having a body with a run line opening, a run insulation piercing member extending into the run line opening, a run guide slot extending from the run line opening, a tap line opening, a tap insulation piercing member extending into the tap line opening, and a tap guide slot extending from the tap line opening; a run cap slidably connected to the conductor block via the run guide slot; a run bolt extending through the run cap into the run line opening, wherein the run bolt includes a first threaded shank and a first shearable head, wherein the first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit; a tap cap slidably connected to the conductor block via the tap guide slot; and a tap bolt extending through the tap cap into the tap line opening, wherein the tap bolt includes a second threaded shank and a second shearable head, wherein the second shearable head is configured to disconnect from the second threaded shank on reaching a second torque limit.

12. The electrical connector of claim 11, wherein the tap opening is a first tap opening, the conductor block further comprises a second tap opening, and the run opening is positioned between the first tap opening and the second tap opening.

13. The electrical connector of claim 11, wherein the conductor block has a substantially L-shaped configuration with the body having a run portion and a tap portion extending from the run portion.

14. The electrical connector of claim 11, wherein the tap opening is a first tap opening, the conductor block further comprises a second tap opening, and the run opening is positioned above the first tap opening and the second tap opening.

15. An electrical connector for connecting a plurality of conductors comprising: a conductor block having a body with a run line opening, a run insulation piercing member extending into the run line opening, a tap line opening, and a tap insulation piercing member extending into the tap line opening; a run cap extending over the run line opening; a run bolt extending through the run cap into the run line opening, wherein the run bolt includes a first threaded shank and a first shearable head, wherein the first shearable head is configured to disconnect from the first threaded shank on reaching a first torque limit; and a tap bolt extending through the conductor block into the tap line opening.

16. The electrical connector of claim 15, wherein the run insulation piercing member includes a set of teeth.

17. The electrical connector of claim 15, wherein the run bolt includes a pressure bar connected to the first shank.

18. The electrical connector of claim 17, wherein the pressure bar includes a second insulation piercing member.

19. The electrical connector of claim 15, wherein the tap opening is a first tap opening, the conductor block further comprises a second tap opening, and the run opening is positioned between the first tap opening and the second tap opening.

20. The electrical connector of claim 15, wherein the conductor block has a substantially round configuration.
